

H47 Gap-and-Go Long

Buying NQ after a large overnight gap up that the first 15 minutes of the cash session confirms: research report with fixed-risk sizing (cohort member #4, Cfa444b4386db8d33)

Prepared for the investment committee · 23 September 2026 · data through 10 August 2026 · all figures per 1 NQ contract (\$20/point) unless stated

Recommendation: SECOND CANDIDATE. Forward-test it next to H41, at small fixed risk, in the same account. It has the highest profit factor of the 25 frozen strategies (2.03 after the freeze) and was positive in every research stage. It had the second-strongest untouched holdout (t 2.18, null p 0.020), and 2026 was SUPPORTED. It barely moves with H41 (correlation about 0.1). The catch is sample size. It trades only about 33 times a year, so the post-freeze record is 119 trades, and a few big days carry the result. Plan for roughly half the recent edge, and expect some flat years.

Executive summary

- **What it is.** On days when NQ's cash session opens at least $0.25 \times D$ above the previous day's close (D = the average daily range of the last 10 sessions), it waits until 09:45. If price is still at least $0.05 \times D$ above the open, it buys. The stop is $1 \times \text{ATR}$. 25% of the position comes off at $+1R$, and the rest runs to 16:00. Long only, at most one trade a day. **About 33 trades a year: 0.6 a week, and a trade in about half of all weeks.**
- **Evidence ladder.** Discovery 2010-18: +\$8.4k, 228 trades, PF 1.49, t 2.08, all three sub-periods positive. Confirmation 2019-22: +\$24.9k, 137 trades, PF 1.77, t 2.20, null p 0.020, all four years positive, tier STRONG. **Untouched holdout 2023-25: +\$29.1k, 99 trades, PF 1.90, t 2.18, null p 0.020, SUPPORTED.** 2026 audit (Jan-Aug): +\$16.7k, 20 trades, PF 2.35, SUPPORTED.
- **After the freeze** (2023-Aug 2026): 119 trades, **PF 2.03**, +\$45.7k on 1 NQ, +61R, +0.51R per trade, t 2.3 in R. Under stress costs it is still +\$44.3k (PF 1.97).
- **How it makes money.** 72% of trades lose. Most are stopped out within minutes (median 10 minutes), averaging $-0.92R$, which is smaller than $-1R$ because the partial sometimes fills first. The 29% that survive to the close average $+4.0R$. The average winner is $5.3 \times$ the average loser.
- **Main weaknesses.** (1) Small sample: 119 post-freeze trades, and after correcting for the whole search the confirmation result is not significant (family-wise p 0.64). (2) Extreme tail dependence: the 10 best post-freeze trades made 103% of net profit. (3) Long only. The mirror-image short rule was break-even, so some of the edge may be Nasdaq's upward drift. (4) Slow to evaluate: judging it properly forward takes years, not months.
- **Fixed risk (new section).** It works cleanly from \$150 per trade upward: no trade is skipped even at 2026 volatility. The profit factor falls from about 2.0 to about 1.8 because its best trades came on volatile gap days. Alone it is too slow for most prop evaluations. Next to H41 in one account it improves both the pass rate and the speed.

Headline results by research stage

Period	Trades	Per week	Net P&L	PF	Win %	Net R	Avg R	Max DD	Verdict
Discovery 2010-18	228	0.5	\$8,416	1.49	28.1%	+109.5	+0.480	\$3,339	PRIMARY
Confirmation 2019-22	137	0.7	\$24,860	1.77	28.5%	+76.3	+0.557	\$5,027	STRONG
Holdout 2023-25	99	0.6	\$29,058	1.90	27.3%	+45.8	+0.463	\$4,621	SUPPORTED
2026 audit (Jan-Aug)	20	0.6	\$16,650	2.35	30.0%	+15.0	+0.751	\$3,032	SUPPORTED
2024-2025 only	68	0.7	\$25,088	2.12	29.4%	+37.8	+0.556	\$3,584	-
All post-freeze 2023-26	119	0.6	\$45,708	2.03	27.7%	+60.8	+0.511	\$4,621	-

Stages before the freeze (discovery, confirmation) were used to select the strategy; 2023 onward was never seen before the cohort was frozen on 22 Sep 2026 23:21 UTC. Verdicts are the campaign's preregistered classifications. R = P&L divided by the trade's own initial risk.

1. How this strategy was found and tested

The strategy came out of the QuantLab V3 research campaign: 53 preregistered hypothesis families (33,077 specifications) were searched on 2010–2018 NQ/ES 1-minute data. Survivors were frozen and re-tested unchanged on 2019–2022 against 100 synthetic ‘no-edge’ market worlds, clustered into distinct behaviours, and 25 were selected into a frozen cohort. The cohort was then run once on the untouched 2023–2025 holdout, and once on 2026 data to 10 Aug 2026. Nothing was changed after any later stage was seen; that is what makes the post-2022 numbers honest out-of-sample evidence.

Stage	Dates	Role	What could change
Discovery	2010-06-07 → 2018-12-31	search (in-sample)	everything: rules, grid, management
Confirmation	2019 → 2022	out-of-sample test vs null worlds	only which frozen candidates are kept
Final cohort freeze	22 Sep 2026 23:21 UTC	spec hash f5666dac...	nothing after this point
Holdout	2023 → 2025	untouched out-of-sample	nothing (report only)
2026 audit	2026-01-01 → 2026-08-10	secondary, low power	nothing (report only)
True forward	from 2026-09-24	live-market paper record	nothing (forward/run_forward.py)

Reproducibility. For this report every trade from 2010 to 2026 was regenerated with the frozen specification, the stage-gated data loader and the campaign engine. Trade counts and net P&L match the sealed campaign records exactly in all four periods for every strategy covered here.

Why R-multiples matter here. NQ traded near 2,000 in 2010 and near 27,000 in 2026, and the stop is volatility-based, so one trade's dollar risk ranged from roughly \$50 to over \$2,000 on one contract. Dollar totals are therefore dominated by recent, high-volatility years, and a single contract is a very different bet in 2012 than in 2026. We report both dollars (the campaign's preregistered metric) and R ($\text{P\&L} \div \text{initial risk}$), which approximates a book that risks a constant amount per trade. **The official verdicts use dollars; the R view is a descriptive, post-hoc lens and cannot overturn them.**

Hypothesis family

Preregistered hypothesis (H47). The gap (RTH open minus the previous RTH close) is at least $g \times D$. Confluence ('open'): at 09:30 + w minutes, the close is at least $0.05 \times D$ beyond the RTH open in the gap's direction. The trade is a continuation, entered at 09:30 + w . Grid: $g \in \{0.25, 0.5\}$, $w \in \{5, 15\}$, open $\in \{\text{none, same}\}$, long / short / both. In discovery 648 specifications were tested, 122 were PRIMARY, and the best t was 2.4.

Rationale. A large overnight gap means information arrived outside the cash session: earnings, macro data, or moves in Asia and Europe. The first minutes of the cash session show whether local participants accept the new price. If price holds above the open, the gap is being accepted, and institutions that must trade in the cash session keep buying for the rest of the day. If price falls back through the open, the gap often fills instead. The confirmation filter separates the two.

The confirmation filter carries the edge. The same gap-up trade without the 'open holds' condition had PF 1.14 in discovery, 1.30 in confirmation, 0.98 in the holdout and 1.07 in 2026. With it: 1.49 / 1.77 / 1.90 / 2.35. Across the family, the condition raised the average trade in 77% of 300 discovery pairs and 61% of 114 confirmation pairs.

Behaviour cluster. Cluster #11: STRONG, 24 confirmation survivors, 3 entry rules. Its siblings in the cohort are #22 (the same idea confirmed at 09:35; holdout correlation 0.60) and #23 (both directions). A cautionary sibling is #17: gap $\geq 0.5D$ with no 'open holds' filter. It FAILED the holdout.

2. H47 (#4, long only): specification

Field	Frozen value
Strategy ID	Cfa444b4386db8d33
Entry rule ID	E8f8a54882d6f9ad0
Family / version	v3_h47_overnight_same_dir_open (v3.0)
Instrument	NQ (E-mini Nasdaq-100 futures, \$20 per point), front contract, 1-minute bars
Direction	long
Mode	continuation
Long trigger	at 09:30+15min, the RTH open gapped $\geq 0.25D$ above the previous RTH close and the close is $\geq 0.05D$ above the RTH open
Short trigger	not traded
Parameters	{"g": 0.25, "mode": "continuation", "open": "same", "w": 15}
Signal timing	evaluated at the close of each 1-minute bar
Entry order	market order at the open of the next 1-minute bar (same session and contract)
Entry window	09:30–15:30 New York; one position at a time; no pyramiding
Initial stop	$1.0 \times ATR_h$ from entry ($ATR_h = ATR(14)$ of completed 15-minute bars); stop order, gaps through the stop fill at the bar open; rounded away from entry to the tick
Target	25% at +1R + 75% runs to the time exit
Break-even / trailing / partials	none / none / 25% / 75%
Time exit	flat at the close of the last bar before 16:00 New York
Same-bar ambiguity	pessimistic (stop assumed first)
Costs (baseline)	\$14 per round trip = \$4 commission + 1 tick slippage per side; also scored at \$0 / \$9 / \$26
Cost file version	costs_v2_approved_2026-09-21#a27ed5df0c04

Statistical evidence trail

Stage	Test / statistic	Result	Verdict
Discovery 2010–18	label (trades ≥ 30 , net > 0 , $t \geq 1$)	228 trades, PF 1.49, t 2.08	PRIMARY
Discovery	parameter robustness (share of grid neighbours positive)	1.00	
Discovery	sub-periods positive (of 3) / years positive (of 9)	3 / 7	
Confirmation 2019–22	t-stat of mean trade	2.20	
Confirmation	null p (zero-drift worlds) / (drift worlds) / max	0.020 / 0.020 / 0.020	$p \leq 0.05$
Confirmation	family-wise p (max-t in same frequency bucket)	0.644	not significant
Confirmation	parameter support / management support	0.50 / 0.54	
Confirmation	years positive (of 4); stress net	4; \$23,216	
Confirmation	evidence tier (preregistered)		STRONG
Behaviour cluster	cluster at p 0.50 (members / entry rules)	#11 (24 / 3)	STRONG
Holdout 2023–25	t-stat; z vs confirmation mean trade	2.18; 0.83	
Holdout	null p (100 holdout null worlds)	0.020	
Holdout	classification (preregistered)		SUPPORTED
2026 audit	t-stat; net	1.25; \$16,650	SUPPORTED

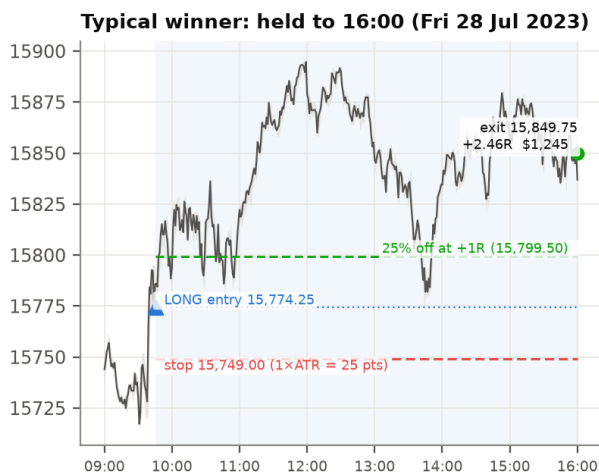
Null worlds randomly flip the sign of each minute's moves (keeping volatility, seasonality, gaps and the NQ-ES relation) so any directional edge is destroyed; p is the share of such worlds that match the real t-stat. The family-wise p corrects for having searched many candidates of similar frequency.

H47 (#4, long only): how it trades

Signal (checked once a day). At the 09:45 bar close, two conditions must hold. (1) Today's regular session opened at least $0.25 \times D$ above yesterday's 16:00 close (D = the mean high-low range of the last 10 regular sessions). (2) Price is still at least $0.05 \times D$ above today's open. If both hold, it buys at market on the next 1-minute bar. On other days it does nothing.

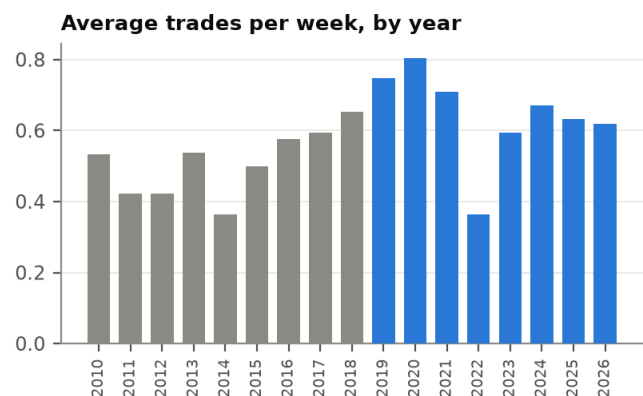
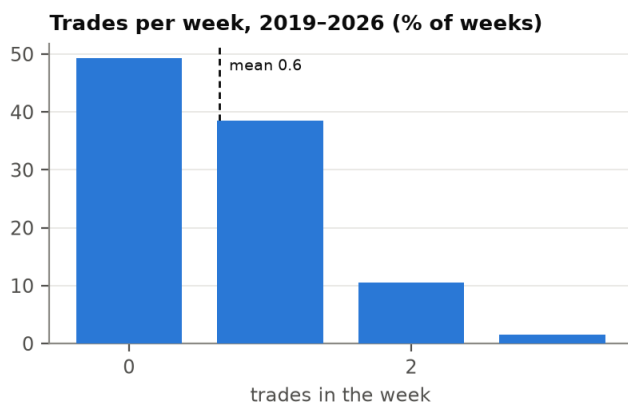
Risk and exit. The stop sits $1 \times \text{ATR}$ below the entry (ATR(14) of completed 15-minute bars): about 22 NQ points in 2023, and 54 points in 2026. One NQ contract therefore risked about \$1,000 in 2026; one MNQ about \$100. A quarter of the position is sold at $+1R$. The other three quarters have no target, no trailing stop and no move to break-even, and are sold at the last bar before 16:00.

What a trade looks like. Failures are quick: the median stopped trade lasts 10 minutes. Winners last all day: the median trade held to the close lasts 375 minutes. About 54% of trades reach $+1R$ and bank the partial, but only 29% finish in profit. The strategy relies on trend days that begin with a gap. This is descriptive only, and the rules are not changed because of it.



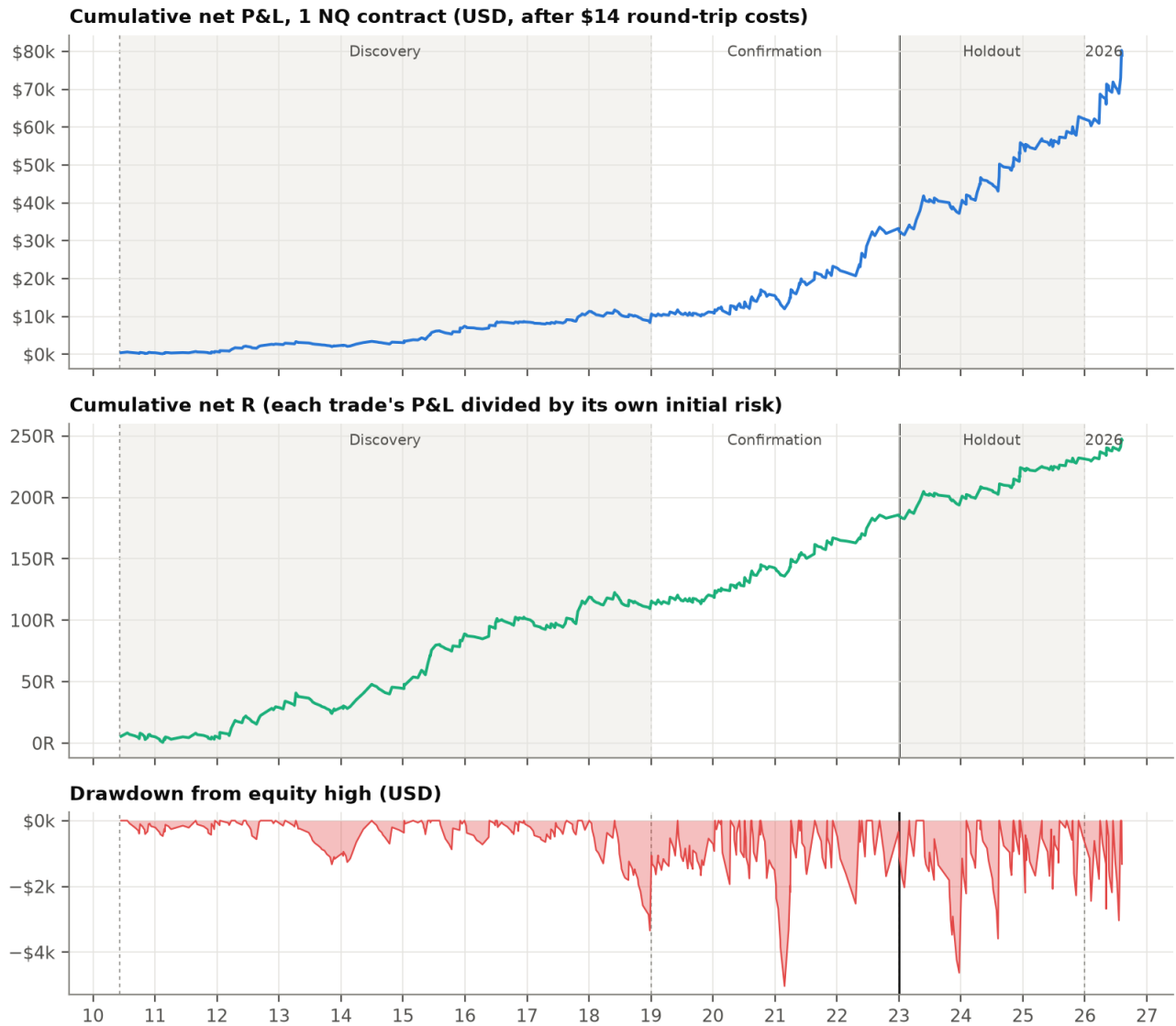
Real holdout trades, 1-minute NQ bars (grey band = bar range, line = close). Blue dotted = entry price, red dashed = the fixed stop; nothing else is managed.

How often it trades



Frequency (2019-Aug 2026)	Value
Average trades per week	0.6
Median trades per week	1
Weeks with no trade	49.4%
Weeks with exactly 1 trade	38.5%
Weeks with 2 or more trades	12.1%
Busiest week	3 trades
Longest gap without a trade	9 weeks
Average trades per year	34

H47 (#4, long only): performance



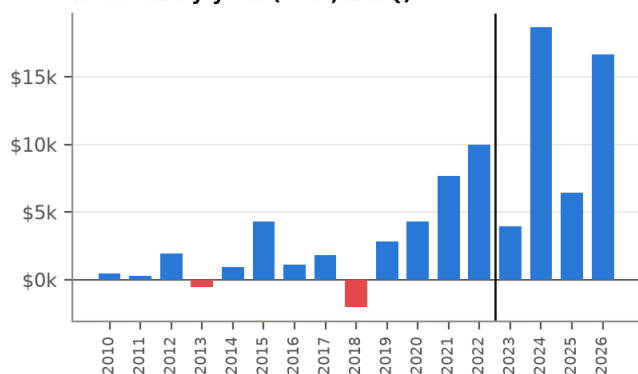
H47 (#4, long only): full statistics by period

Metric	Discovery 2010-18	Confirm. 2019-22	Holdout 2023-25	2026 Jan-Aug	2024-25	Post-freeze 2023-26	Full 2010-26
Trades	228	137	99	20	68	119	484
Trades / year	27	34	33	32	34	33	30
Trades / week (all weeks)	0.5	0.7	0.6	0.6	0.7	0.6	0.6
Weeks with at least 1 trade	45%	52%	49%	50%	49%	49%	48%
Trades / week (active weeks)	1.1	1.3	1.3	1.2	1.3	1.3	1.2
Days with a trade (of sessions)	11%	14%	13%	13%	14%	13%	12%
Max trades in one day	1	1	1	1	1	1	1
Long share	100%	100%	100%	100%	100%	100%	100%
Net P&L (baseline \$14)	\$8,416	\$24,860	\$29,058	\$16,650	\$25,088	\$45,708	\$78,983
Gross P&L (no costs)	\$11,608	\$26,778	\$30,444	\$16,930	\$26,040	\$47,374	\$85,759
Net P&L, moderate \$9	\$9,556	\$25,544	\$29,553	\$16,750	\$25,428	\$46,303	\$81,403
Net P&L, stress \$26	\$5,680	\$23,216	\$27,870	\$16,410	\$24,272	\$44,280	\$73,175
Profit factor (net)	1.49	1.77	1.90	2.35	2.12	2.03	1.84
Profit factor (gross)	1.78	1.87	1.98	2.39	2.20	2.09	1.96
Profit factor (stress)	1.30	1.69	1.84	2.31	2.06	1.97	1.75
Win rate	28.1%	28.5%	27.3%	30.0%	29.4%	27.7%	28.1%
Average trade	\$37	\$181	\$294	\$832	\$369	\$384	\$163
Average winner	\$398	\$1,464	\$2,268	\$4,832	\$2,376	\$2,734	\$1,271
Average loser	-\$104	-\$329	-\$447	-\$882	-\$467	-\$518	-\$270
Payoff ratio (avg W / avg L)	3.83	4.45	5.07	5.48	5.09	5.28	4.71
Expectancy (net R / trade)	+0.480	+0.557	+0.463	+0.751	+0.556	+0.511	+0.510
Total net R	+109.5	+76.3	+45.8	+15.0	+37.8	+60.8	+246.6
Median trade (R)	-0.72	-0.58	-0.53	-0.78	-0.53	-0.53	-0.69
t-stat of mean trade (USD)	2.08	2.20	2.18	1.25	2.15	2.44	3.53
t-stat of mean trade (R)	2.78	2.63	1.96	1.24	1.99	2.32	4.45
Sharpe, daily, annualised (USD)	0.69	1.07	1.22	1.54	1.46	1.24	0.85
Sharpe, daily, annualised (R)	0.92	1.27	1.10	1.53	1.36	1.18	1.07
Sortino (USD)	1.77	3.17	4.01	5.80	4.91	4.33	2.81
Max drawdown (USD)	\$3,339	\$5,027	\$4,621	\$3,032	\$3,584	\$4,621	\$5,027
Max drawdown (R)	16.7	9.3	11.0	3.1	6.2	11.0	16.7
Longest drawdown (calendar days)	397	216	251	56	103	251	397
Calmar (annual net / max DD)	0.29	1.24	2.10	8.91	3.50	2.73	0.97
Longest losing streak (trades)	18	9	8	4	8	8	18
Longest winning streak	4	2	3	2	2	3	4
Best trade	\$1,097	\$3,805	\$4,951	\$7,708	\$4,951	\$7,708	\$7,708
Worst trade	-\$484	-\$1,189	-\$1,084	-\$1,339	-\$1,084	-\$1,339	-\$1,339
Best day / worst day	\$1,097 / -\$484	\$3,805 / -\$1,189	\$4,951 / -\$1,084	\$7,708 / -\$1,339	\$4,951 / -\$1,084	\$7,708 / -\$1,339	\$7,708 / -\$1,339
Top-10 trades as % of net	98%	110%	113%	164%	124%	103%	61%
Positive months	50% of 94	61% of 46	43% of 35	75% of 8	46% of 24	49% of 43	52% of 183
Average hold (minutes)	142	132	128	122	137	127	135
Avg initial risk (NQ points)	5.4	18.4	28.1	50.3	30.2	31.8	15.6
Avg initial risk, 1 NQ (USD) = 1R	\$108	\$368	\$562	\$1,006	\$603	\$637	\$312
Average NQ price at entry	4,263	11,027	18,917	28,029	21,016	20,448	10,157

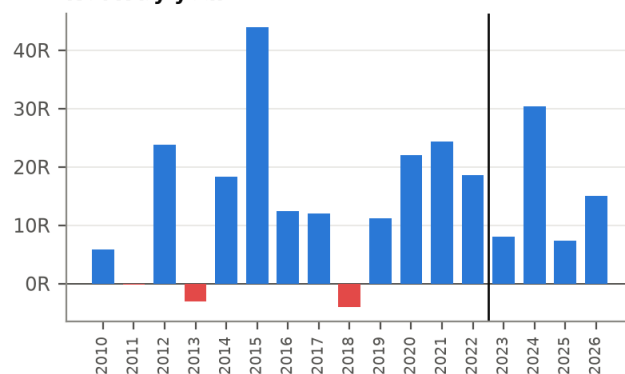
Sharpe and Sortino use every business day in the period (zero on no-trade days) and are scale-free. The strategy is always flat overnight.

H47 (#4, long only): calendar

Net P&L by year (USD, 1 NQ)



Net R by year

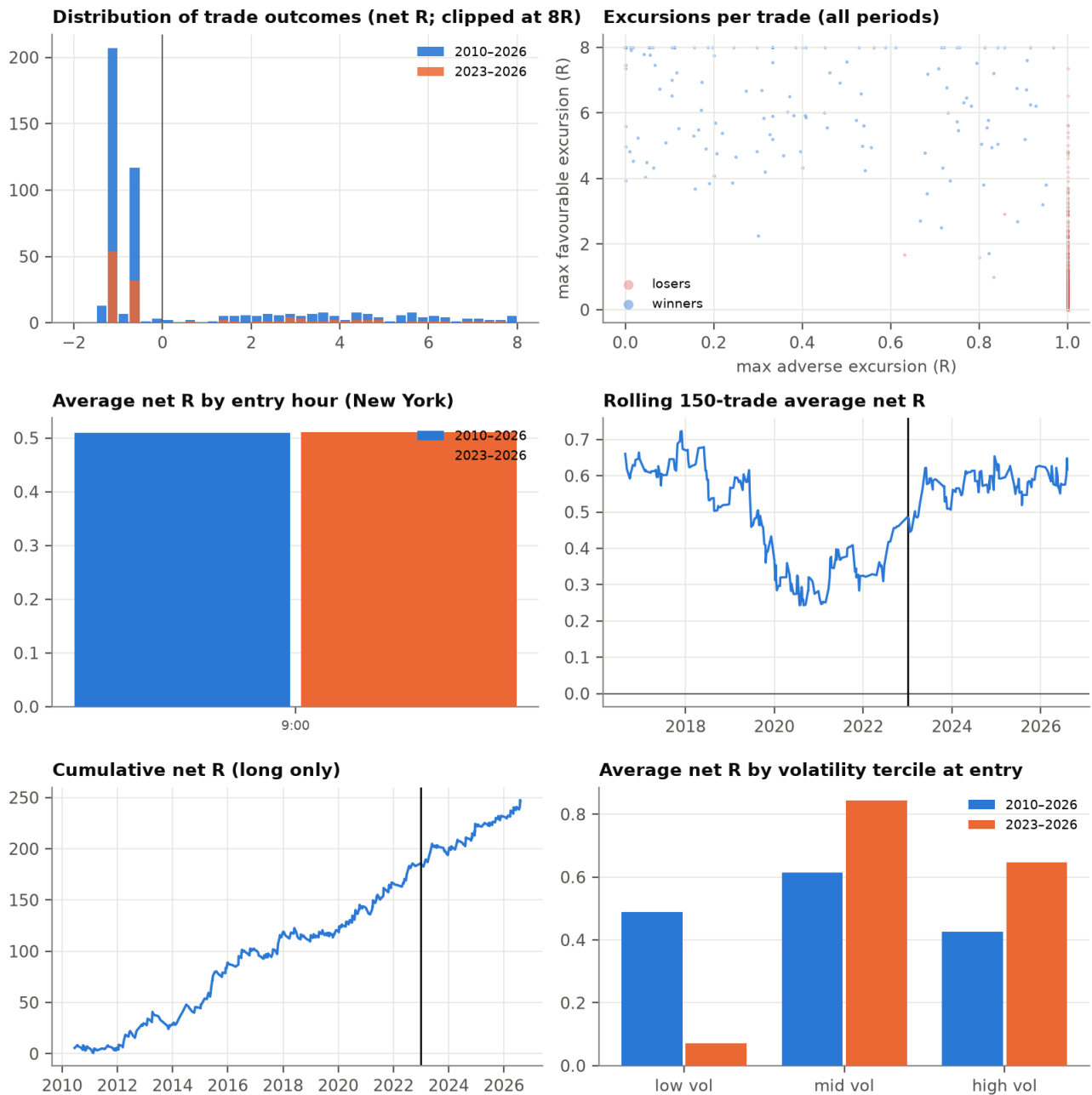


Year	Period	N	Net P&L	Gross	Stress	PF	Win %	Avg trade	Net R	Avg R	Max DD	1R (USD)
2010	Disc.	16	\$475	\$699	\$283	1.61	38%	\$30	+5.9	+0.370	\$410	\$72
2011	Disc.	22	\$303	\$611	\$39	1.26	27%	\$14	-0.1	-0.006	\$430	\$84
2012	Disc.	22	\$1,970	\$2,278	\$1,706	2.85	41%	\$90	+23.9	+1.085	\$566	\$77
2013	Disc.	28	-\$522	-\$130	-\$858	0.70	11%	-\$19	-3.0	-0.108	\$1,330	\$75
2014	Disc.	19	\$942	\$1,208	\$714	1.79	37%	\$50	+18.4	+0.967	\$702	\$89
2015	Disc.	26	\$4,286	\$4,650	\$3,974	3.47	42%	\$165	+43.9	+1.690	\$875	\$126
2016	Disc.	30	\$1,130	\$1,550	\$770	1.60	30%	\$38	+12.5	+0.416	\$726	\$111
2017	Disc.	31	\$1,851	\$2,285	\$1,479	1.94	26%	\$60	+12.1	+0.390	\$547	\$87
2018	Disc.	34	-\$2,018	-\$1,542	-\$2,426	0.64	15%	-\$59	-4.0	-0.118	\$3,339	\$200
2019	Conf.	39	\$2,823	\$3,369	\$2,355	1.54	28%	\$72	+11.2	+0.286	\$1,662	\$209
2020	Conf.	42	\$4,334	\$4,922	\$3,830	1.42	29%	\$103	+22.0	+0.524	\$1,933	\$370
2021	Conf.	37	\$7,711	\$8,229	\$7,267	1.80	27%	\$208	+24.4	+0.660	\$3,526	\$406
2022	Conf.	19	\$9,992	\$10,258	\$9,764	2.43	32%	\$526	+18.7	+0.983	\$2,522	\$613
2023	Holdout	31	\$3,970	\$4,404	\$3,598	1.41	23%	\$128	+8.1	+0.260	\$4,621	\$472
2024	Holdout	35	\$18,668	\$19,158	\$18,248	2.78	31%	\$533	+30.5	+0.870	\$3,584	\$572
2025	Holdout	33	\$6,420	\$6,882	\$6,024	1.54	27%	\$195	+7.3	+0.222	\$2,271	\$637
2026	Audit	20	\$16,650	\$16,930	\$16,410	2.35	30%	\$832	+15.0	+0.751	\$3,032	\$1,006

Monthly net R, 2019-2026 (blue = gain, red = loss; blank = no data)

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
2019	+3.5	+1.4	+0.2	+4.2	-2.1	-0.5	+0.3	-1.3	+2.1	-0.5	+2.5	+1.4
2020	+1.8	+2.6		+3.7	-0.1	-0.3	+3.3	+5.3	-0.5	+7.8	-0.5	-1.1
2021	-4.2	-2.6	+7.8	+4.4	+4.3	-0.5	-1.6	+11.4	-2.2	-2.1	+4.1	+5.5
2022	-2.1	-0.5	-0.5	-1.0	+7.6	+4.2	+8.4	-2.1	+3.6	-1.5		+2.7
2023	-2.6	-0.5	+4.3	+4.9	+12.0	-0.8	+0.4	-1.6		-2.6	-3.4	-2.1
2024	+6.2	+0.3	-1.1	+9.4	-1.1	-1.1	-3.1	+7.0	-0.5	-2.1	+6.1	+10.3
2025	-1.1	-1.0	-0.5	+3.1	-1.0	-1.5	+1.0	+3.2	+3.2	-0.7	+3.2	-0.5
2026	-1.0	+1.8	+4.8	-1.0	+2.1	+1.5	+1.4	+5.4				

H47 (#4, long only): trade anatomy



Volatility tercile = initial stop distance as a share of price, ranked within each calendar year.

H47 (#4, long only): breakdowns (all periods vs post-freeze 2023-26)

Exit reason

Exit	Trades (all)	Avg R (all)	PF (all)	Net (all)	Trades 23-26	Avg R 23-26	PF 23-26	Net 23-26
16:00 flat	140	+4.009	2,669.28	\$172,771	33	+4.033	-	\$90,237
Stop-loss hit	344	-0.915	0.00	-\$93,788	86	-0.840	0.00	-\$44,529

Entry hour (New York)

Entry hour	Trades (all)	Avg R (all)	PF (all)	Net (all)	Trades 23-26	Avg R 23-26	PF 23-26	Net 23-26
09:00-09:59	484	+0.510	1.84	\$78,983	119	+0.511	2.03	\$45,708

Weekday

Weekday	Trades (all)	Avg R (all)	PF (all)	Net (all)	Trades 23-26	Avg R 23-26	PF 23-26	Net 23-26
Mon	108	+0.715	1.98	\$19,506	30	+0.601	2.09	\$10,869
Tue	113	+0.550	2.33	\$23,352	21	+0.810	3.67	\$19,280
Wed	88	+0.226	0.96	-\$764	18	-0.044	0.80	-\$1,632
Thu	80	+0.230	1.57	\$8,511	18	+0.466	2.33	\$7,919
Fri	95	+0.726	2.34	\$28,379	32	+0.569	1.71	\$9,272

H47 (#4, long only): robustness

Parameter neighbourhood. Every grid point of the family with the same stop and exit, both directions separately. If the frozen point is a lucky outlier, its neighbours will look much worse. Rerun post hoc for this report; purely descriptive, and it does not re-select anything.

Entry parameters	Dir.	Disc. n / PF	Conf. n / PF	Holdout n / PF	Holdout net	2026 n / PF	2026 net
g=0.25, w=15, open=same ◀ frozen	long	228 / 1.49	137 / 1.77	99 / 1.90	\$29,058	20 / 2.35	\$16,650
g=0.25, w=15, open=same	short	154 / 1.24	101 / 1.16	66 / 1.00	-\$116	15 / 1.02	\$292
g=0.25, w=15, open=same	both	382 / 1.38	238 / 1.45	165 / 1.41	\$28,941	35 / 1.65	\$16,942
g=0.25, w=5, open=same	long	211 / 1.21	118 / 1.78	89 / 1.22	\$6,239	19 / 2.11	\$11,654
g=0.25, w=5, open=same	short	144 / 1.06	90 / 1.59	61 / 1.97	\$22,450	15 / 0.00	-\$13,025
g=0.25, w=5, open=same	both	355 / 1.14	208 / 1.68	150 / 1.55	\$28,689	34 / 0.94	-\$1,371
g=0.25, w=15, open=None	long	664 / 1.14	334 / 1.30	229 / 0.98	-\$2,020	52 / 1.07	\$2,842
g=0.25, w=15, open=None	short	468 / 1.19	254 / 0.99	180 / 0.88	-\$11,544	42 / 0.85	-\$5,529
g=0.25, w=15, open=None	both	1132 / 1.16	588 / 1.15	409 / 0.93	-\$13,564	94 / 0.96	-\$2,687
g=0.25, w=5, open=None	long	664 / 0.96	334 / 1.20	229 / 1.09	\$6,795	52 / 1.24	\$7,320
g=0.25, w=5, open=None	short	468 / 0.72	253 / 1.16	180 / 1.01	\$1,036	42 / 0.80	-\$6,219
g=0.25, w=5, open=None	both	1132 / 0.84	587 / 1.18	409 / 1.05	\$7,832	94 / 1.02	\$1,100
g=0.5, w=15, open=same	long	104 / 1.78	63 / 1.28	46 / 1.18	\$2,977	14 / 3.60	\$19,613
g=0.5, w=15, open=same	short	74 / 0.86	47 / 1.09	39 / 0.88	-\$3,130	12 / 1.33	\$3,462
g=0.5, w=15, open=same	both	178 / 1.32	110 / 1.18	85 / 1.00	-\$152	26 / 2.28	\$23,075
g=0.5, w=5, open=same	long	96 / 1.23	57 / 1.43	42 / 1.01	\$123	11 / 2.93	\$10,018
g=0.5, w=5, open=same	short	71 / 0.53	44 / 1.37	34 / 2.39	\$18,819	12 / 0.00	-\$9,346
g=0.5, w=5, open=same	both	167 / 0.90	101 / 1.39	76 / 1.72	\$18,942	23 / 1.05	\$673
g=0.5, w=15, open=None	long	283 / 1.37	153 / 1.43	121 / 0.70	-\$14,755	35 / 1.16	\$4,499
g=0.5, w=15, open=None	short	234 / 0.86	130 / 0.98	94 / 1.04	\$2,179	30 / 1.14	\$3,509
g=0.5, w=15, open=None	both	517 / 1.09	283 / 1.18	215 / 0.88	-\$12,576	65 / 1.15	\$8,008
g=0.5, w=5, open=None	long	283 / 0.98	153 / 1.11	121 / 0.69	-\$14,090	35 / 1.71	\$12,810
g=0.5, w=5, open=None	short	234 / 0.51	129 / 0.80	94 / 1.29	\$12,722	30 / 1.00	-\$78
g=0.5, w=5, open=None	both	517 / 0.73	282 / 0.93	215 / 0.98	-\$1,369	65 / 1.32	\$12,732

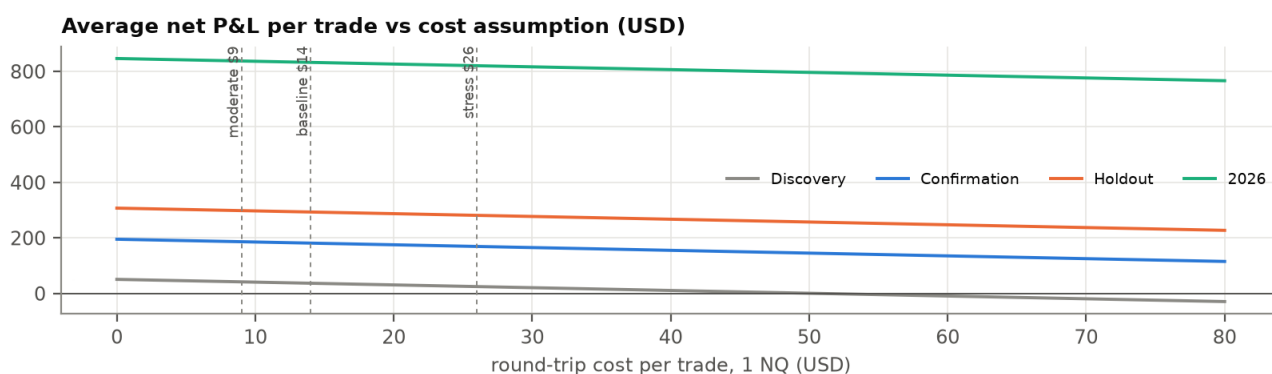
Neighbours with ≥ 20 trades and positive net: Discovery 14/24; Confirmation 20/24; Holdout 14/24; 2026 12/17

Stop / exit management. The frozen entry under every management variant from the campaign library (discovery stages A and B).

Stop / exit management	Disc. PF	Conf. PF	Holdout n / PF	Holdout net	2026 PF	2026 net
1×ATR stop; single target at +1R	0.84	0.95	99 / 1.32	\$7,754	0.90	-\$1,065
1×ATR stop; single target at +1.5R	0.97	1.05	99 / 1.36	\$10,479	0.82	-\$2,325
1×ATR stop; single target at +2R	1.01	1.17	99 / 1.58	\$17,959	0.89	-\$1,570
1×ATR stop; single target at +3R	1.22	1.36	99 / 1.63	\$22,464	1.34	\$4,705
1×ATR stop; single target at +4R	1.25	1.62	99 / 1.75	\$28,679	1.76	\$10,665
1×ATR stop; no target, flat 16:00	1.59	1.88	99 / 1.90	\$36,159	2.61	\$22,555
1×ATR stop; no target, exit after 60 min	1.37	1.62	99 / 1.59	\$20,454	0.85	-\$2,100
1.5×ATR stop; single target at +1R	0.82	1.06	99 / 1.04	\$1,804	0.83	-\$2,815
1.5×ATR stop; single target at +2R	1.02	1.32	99 / 1.27	\$13,764	1.37	\$6,715
1.5×ATR stop; no target, flat 16:00	1.34	1.70	99 / 1.46	\$26,124	2.58	\$28,260
5-min swing stop; single target at +1R	1.09	1.12	99 / 0.98	-\$1,521	0.80	-\$4,725
5-min swing stop; single target at +2R	1.22	1.23	99 / 1.26	\$19,439	1.20	\$4,890
5-min swing stop; no target, flat 16:00	1.42	1.43	99 / 1.25	\$19,269	1.76	\$18,485
1×ATR stop; stop to entry at +0.5R, target +2R	0.37	0.60	99 / 1.11	\$1,794	0.51	-\$2,565
1×ATR stop; stop to entry at +0.5R, no target	0.49	1.00	99 / 1.84	\$13,189	0.00	-\$5,215
1×ATR stop; stop to entry at +1R, target +2R	0.82	1.04	99 / 1.21	\$5,214	1.01	\$100
1×ATR stop; stop to entry at +1R, no target	1.21	1.87	99 / 1.73	\$17,809	1.34	\$3,655
1×ATR stop; stop to entry at +1.5R, target +2R	0.98	1.15	99 / 1.47	\$13,789	0.95	-\$620
1×ATR stop; stop to entry at +1.5R, no target	1.55	1.94	99 / 2.29	\$38,454	1.61	\$8,065
1×ATR stop; no target, 1.5×ATR trailing stop	1.47	1.36	99 / 1.56	\$14,404	1.47	\$4,910
1×ATR stop; no target, 2×ATR trailing stop	1.57	1.87	99 / 1.93	\$28,254	1.43	\$5,590
1×ATR stop; no target, 3×ATR trailing stop	1.63	1.66	99 / 1.97	\$35,184	2.61	\$22,555

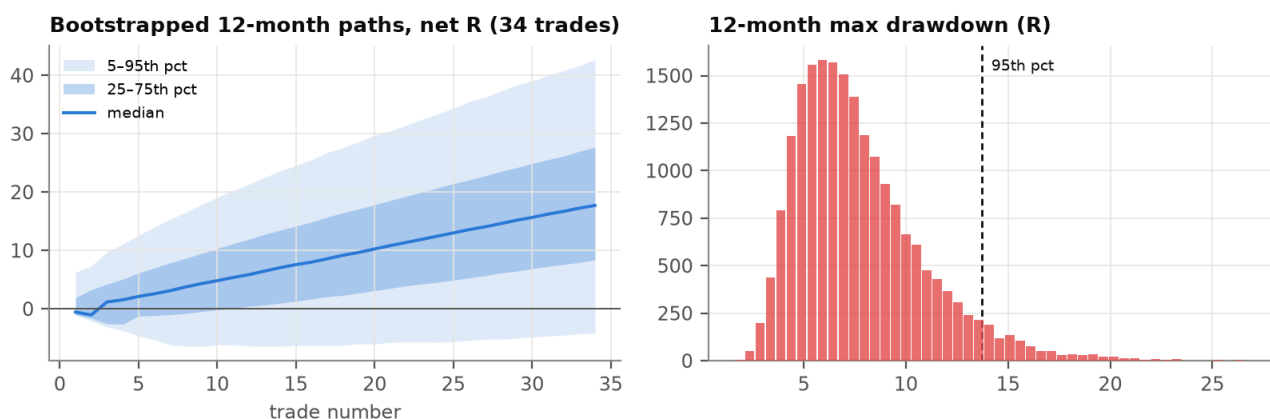
Stop / exit management	Disc. PF	Conf. PF	Holdout n / PF	Holdout net	2026 PF	2026 net
1×ATR stop; no target, trail beyond last 5-min swing	1.56	1.85	99 / 2.01	\$34,559	1.64	\$8,370
1×ATR stop; 50% at +1R, 50% at +2R	0.93	1.08	99 / 1.53	\$12,856	0.88	−\$1,318
1×ATR stop; 50% at +1R, 50% at +3R	1.07	1.22	99 / 1.62	\$15,109	1.17	\$1,820
1×ATR stop; 50% at +1R, rest trails 2×ATR	1.27	1.48	99 / 1.75	\$18,004	1.21	\$2,262
1×ATR stop; 25% at +1R, 75% runs to 16:00 ◀ frozen	1.49	1.77	99 / 1.90	\$29,058	2.35	\$16,650
1×ATR stop; 50% at +1R then stop to entry, rest +3R	0.95	1.07	99 / 1.44	\$10,419	0.81	−\$2,050
1×ATR stop; stop to entry at +1R, 2×ATR trail	1.10	1.61	99 / 1.85	\$20,689	1.10	\$1,020

Cost sensitivity. The line shows what the average trade would be at any round-trip cost.



Break-even round-trip cost per contract (gross average trade): Discovery \$51; Confirmation \$195; Holdout \$308; 2026 \$846. The baseline assumption is \$14.

H47 (#4, long only): risk, sizing and forward expectations



Bootstrap of 256 trades from 2019–Aug 2026 (all out-of-sample relative to discovery), 34 trades per simulated year, 20,000 paths, trades drawn independently. This ignores regime persistence, so real drawdowns cluster more than shown. It assumes the 2019–26 edge continues unchanged; the section 'Will it perform like this in the future?' repeats it with smaller edges.

Forward 12-month statistic (bootstrap)	In R	1 NQ at 2026 risk (1R ≈ \$1,006)	1 MNQ (1R ≈ \$101)
Median 12-month result	+17.7	\$17,807	\$1,781
5th percentile (bad year)	−4.2	−\$4,225	−\$423
95th percentile (good year)	+42.7	\$42,926	\$4,293
Probability of a losing year	10%		
Median max drawdown	6.9	\$6,967	\$697
95th pct max drawdown	13.7	\$13,792	\$1,379
99th pct max drawdown	17.9	\$18,004	\$1,800
Historical max drawdown 2010–26	16.7		

Suggested capital rule of thumb (not a margin requirement): hold at least the 99th-percentile 12-month drawdown plus exchange margin per unit traded. Dollar outcomes on one full NQ contract are far larger today than in most of the backtest; micro contracts (MNQ, 1/10 size) allow risk to be held near-constant per trade.

Portfolio context (25-strategy frozen cohort, 1 contract each)

Period	Corr. with rest of cohort (daily)	Median pairwise corr.	Most correlated members	Share of cohort P&L
holdout	0.23	0.04	#22 H47 long (0.60), #23 H47 both (0.50), #11 H07 long (0.40), #17 H47 long (0.38)	5%
2026	0.48	0.21	#17 H47 long (0.87), #22 H47 long (0.57), #23 H47 both (0.56), #11 H07 long (0.52)	17%

Strengths and weaknesses

Strengths

- **Consistent across 17 years.** 14 of 17 calendar years were positive in R. All three discovery sub-periods and all four confirmation years were positive. Unlike H41, it did not depend on one era.
- **Clean post-selection evidence.** Holdout null p 0.020 and t 2.18, the second best in the cohort. 2026 was SUPPORTED as well. The post-freeze average trade (+0.51R) matched discovery (+0.48R) and confirmation (+0.56R), with no sign of decay.
- **The highest profit factor in the cohort.** PF 2.03 after the freeze, 1.97 under stress costs. Costs are negligible next to an average trade of several hundred dollars on 1 NQ.
- **The filter makes the difference, and neighbours agree.** Without the 'open holds' condition the trade is roughly break-even. All four long-only 'open holds' grid points (gap 0.25 or 0.5, confirmed at 09:35 or 09:45) were profitable in both the holdout and 2026. 28 of 29 stop/exit variants were profitable in the holdout.
- **A true diversifier.** Daily correlation after the freeze is about 0.1 with H41, 0.23 with the rest of the cohort, and a median of 0.04 across members. It adds return from a different source: news gaps rather than intraday momentum.
- **Shallow drawdowns in R.** Its worst drawdown ever was 16.7R (11R after the freeze), compared with 123R for H41. The longest losing streak since 2019 is 9 trades.
- **Easy to execute.** One decision a day at a fixed time (09:45), one order, one stop, one partial target. It can be traded by hand.

Weaknesses and risks

- **Small sample.** 119 post-freeze trades and 484 in total. The post-freeze t-stat is 2.3 in R, decent but far from conclusive. The family-wise p of 0.64 says a confirmation result this good was not unusual among everything searched. The holdout and 2026 are what make the case.
- **Extreme tail dependence.** 72% of trades lose, and the 10 best post-freeze trades made 103% of net profit, so all the others together lost money. Missing a single big gap-and-go day can cost a quarter of a year's profit. Execution must never skip a signal.
- **Long only in a long bull market.** The short mirror (gap down, open holds lower) was break-even in the holdout (PF 1.00) and in 2026 (PF 1.02). The edge could partly be Nasdaq's upward drift on strong days. Against that, gap-up longs without the filter did not make money (holdout PF 0.98), so it is not simply 'buy gap-ups'. A prolonged bear market is still untested after the freeze.
- **Slow to judge and slow to earn.** About 33 trades a year. With the planning-case edge, roughly 3 years in 10 lose money even if nothing is wrong, and 150 forward trades would take four to five years. The forward gate must be set on fewer trades.
- **Uneven years.** 2024 made +30.5R, while 2023 (+8.1R) and 2025 (+7.3R) were modest. One good year can carry the record.
- **Sibling risk.** A close cousin (#17, larger gaps, no 'open holds' filter) FAILED the holdout. Small rule changes matter here, so the frozen rule must be run exactly.

Will it perform like this in the future?

Short answer: positive, but noisier and weaker than 2023-26. What argues for persistence: the average trade has been remarkably stable across all four stages (+0.48R, +0.56R, +0.46R, +0.75R), there is a clear economic story, and neighbouring rules agree. What argues for a haircut: it was chosen as the second best of 25 after the holdout, the sample is small, and some of the edge may be bull-market drift. The table resamples its own trades at five assumed average trades.

Scenario for the next 12 months	Avg R / trade	PF (R)	Median 12-mo	Bad year (5th pct)	Good year (95th)	P(losing year)	95th pct max DD
As in the post-freeze record (2023-26)	+0.511	1.77	+17R	-5R	+42R	11%	14R
As in confirmation (2019-22)	+0.557	1.89	+18R	-3R	+43R	9%	13R
Planning case: edge roughly halved	+0.250	1.28	+8R	-14R	+33R	29%	20R
Weak edge	+0.100	1.09	+3R	-19R	+28R	43%	24R
Edge gone	+0.000	0.98	-1R	-22R	+24R	52%	27R

Each row resamples the 2019-2026 trades (34 trades per simulated year) with the average trade shifted to the scenario value. In dollars, multiply R by about \$1,006 for 1 NQ or \$101 for 1 MNQ at 2026 volatility. The highlighted row is the planning case this report recommends.

Our planning case is the highlighted row: about +0.25R per trade, half the post-freeze average. With only about 34 trades a year, that gives a median year of about +8R and roughly a 29% chance of a losing year, even with a real edge. Judge H47 over two to three years, not months. The low frequency is also why it is best run alongside H41 rather than alone.

What would make us more confident: 60 or more forward trades (about two years) at +0.2R or better, with profits not hinging on one or two days. **What would make us stop:** a drawdown beyond 20R (the 95th percentile even in the planning case, and above anything seen since 2010), or 60 forward trades averaging below 0R. The bootstrap assumes independent trades, so real losing years can cluster.

Fixed risk on MNQ instead of fixed contracts

Bottom line. Fixed risk works for H47 at any budget from \$150 up. Even in 2026 one MNQ risked only about \$107 (median stop 54 points), so no trade is skipped from \$150 up. Only \$100 loses trades: 55% of 2026 signals. Unlike H41, fixed risk **lowers** this strategy's profit factor, from 2.03 to about 1.75 after the freeze. Its best trades came on the more volatile gap days, and a fixed contract puts more dollars on exactly those days. The edge remains large either way. The real constraint is frequency: about 33 trades a year makes it slow to pass a prop evaluation on its own. It works better alongside H41.

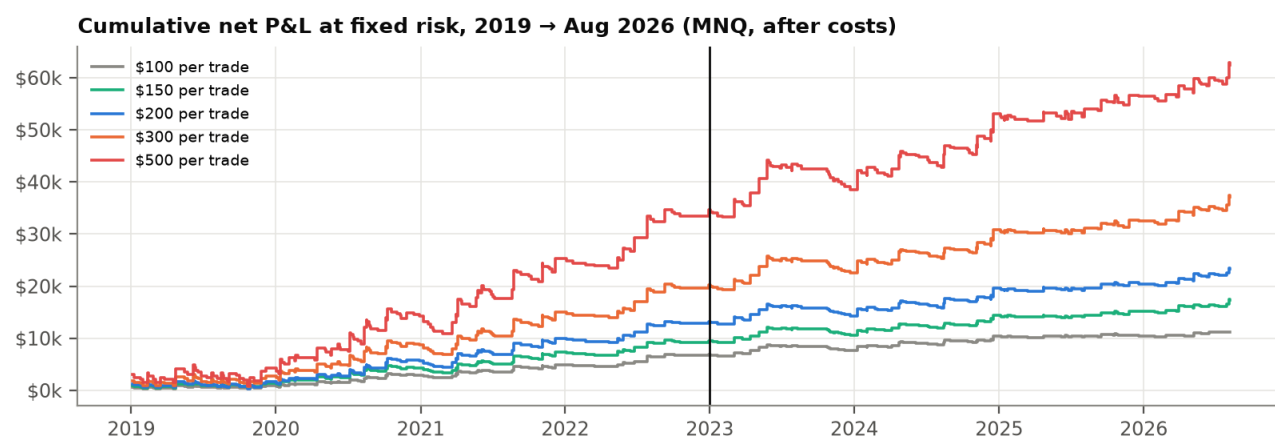
What changes

- **Sizing rule.** Contracts = risk budget ÷ (stop distance in points × \$2), rounded down; no trade if 0. MNQ costs are assumed at \$2.50 per round trip (\$4.25 stress).
- **Profit factor falls, but stays high.** After the freeze, trades on the calmest third of days averaged +0.07R. Mid- and high-volatility days averaged +0.84R and +0.65R. Fixed contracts overweight the big-stop days, which were the good ones, so equalising risk trims PF from about 2.0 to about 1.8. That is still far above H41's.
- **The 25% partial needs at least 3 MNQ.** The frozen plan takes 25% off at +1R. With whole contracts, the partial is rounded to the nearest contract (ties down), so at 1-2 MNQ every contract simply runs to the close. At \$100-\$150 that happens on most 2026 trades. Historically the runner leg alone averaged +0.70R, compared with +0.54R for the frozen plan, so small sizes have not been hurt. That is descriptive only: do not change the rule for larger sizes because of it.
- **Dollar drawdowns are small because trades are rare.** At \$200 per trade the worst post-freeze drawdown was \$2,256 and the worst day −\$212. There is at most one trade a day, always entered at 09:45, so a daily loss limit can never be hit by more than one trade.

Post-freeze 2023 → Aug 2026 (data never used to select the strategy)

Metric	1 NQ fixed	1 MNQ fixed	\$100 risk	\$150 risk	\$200 risk	\$300 risk	\$500 risk
Trades taken (skipped)	119 (0)	119 (0)	104 (15)	119 (0)	119 (0)	119 (0)	119 (0)
Profit factor	2.03	2.04	1.70	1.80	1.75	1.84	1.78
Profit factor, stress costs	1.97	1.98	1.63	1.73	1.68	1.76	1.71
Win rate	28%	28%	27%	28%	28%	28%	28%
Net P&L	\$45,708	\$5,740	\$4,300	\$7,770	\$10,093	\$16,862	\$27,776
Net P&L per year	\$12,661	\$1,590	\$1,191	\$2,152	\$2,796	\$4,671	\$7,694
Average trade	\$384	\$48	\$41	\$65	\$85	\$142	\$233
Average \$ risk per trade	\$637	\$64	\$77	\$121	\$169	\$268	\$468
Risk range (min – max)	\$290 – \$1,430	\$29 – \$143	\$50 – \$100	\$76 – \$150	\$100 – \$200	\$201 – \$300	\$380 – \$500
Contracts (avg / max)	1.0 / 1	1.0 / 1	1.5 / 3	2.2 / 5	3.2 / 6	4.9 / 10	8.6 / 17
Trades that can take the 25% partial	100%	0%	2%	43%	66%	87%	100%
Max drawdown (closed trades)	\$4,621	\$562	\$1,134	\$1,542	\$2,256	\$3,302	\$5,617
Worst losing run	−\$3,584	−\$483	−\$640	−\$908	−\$1,310	−\$1,802	−\$3,186
Worst day	−\$1,339	−\$135	−\$105	−\$158	−\$212	−\$316	−\$536

1 NQ = the frozen strategy exactly. MNQ columns use \$2/point, \$2.50 per round trip per contract, and the partial rounded to whole contracts. Drawdowns use closed-trade equity.



Full out-of-sample record 2019 → Aug 2026

Metric	1 NQ fixed	1 MNQ fixed	\$100 risk	\$150 risk	\$200 risk	\$300 risk	\$500 risk
Trades taken (skipped)	256 (0)	256 (0)	240 (16)	256 (0)	256 (0)	256 (0)	256 (0)
Profit factor	1.92	1.95	1.79	1.79	1.77	1.81	1.79
Profit factor, stress costs	1.85	1.87	1.68	1.68	1.66	1.69	1.68
Win rate	28%	28%	28%	28%	28%	28%	28%
Net P&L	\$70,567	\$8,943	\$11,174	\$17,352	\$23,318	\$37,122	\$62,500
Net P&L per year	\$9,273	\$1,175	\$1,468	\$2,280	\$3,064	\$4,878	\$8,213
Average trade	\$276	\$35	\$47	\$68	\$91	\$145	\$244
Average \$ risk per trade	\$493	\$49	\$80	\$128	\$177	\$276	\$476
Risk range (min - max)	\$105 - \$1,430	\$10 - \$143	\$50 - \$100	\$76 - \$150	\$100 - \$200	\$201 - \$300	\$380 - \$500
Contracts (avg / max)	1.0 / 1	1.0 / 1	2.4 / 9	3.6 / 14	5.0 / 19	7.6 / 28	13.1 / 47
Trades that can take the 25% partial	100%	0%	28%	63%	80%	94%	100%
Max drawdown (closed trades)	\$5,027	\$582	\$1,134	\$1,542	\$2,256	\$3,302	\$5,617
Worst losing run	-\$3,828	-\$483	-\$646	-\$1,006	-\$1,415	-\$2,194	-\$3,916
Worst day	-\$1,339	-\$135	-\$117	-\$182	-\$247	-\$364	-\$611

Stop size and contracts by year

Year	Trades	Median stop (pts)	1 MNQ risk	\$100: MNQ (skip)	\$150: MNQ (skip)	\$200: MNQ (skip)	\$300: MNQ (skip)	\$500: MNQ (skip)
2019	39	10.2	\$20	4 (0%)	7 (0%)	9 (0%)	14 (0%)	24 (0%)
2020	42	17.6	\$35	2 (0%)	4 (0%)	5 (0%)	8 (0%)	14 (0%)
2021	37	19.5	\$39	2 (3%)	3 (0%)	5 (0%)	7 (0%)	12 (0%)
2022	19	31.8	\$64	1 (0%)	2 (0%)	3 (0%)	4 (0%)	7 (0%)
2023	31	22.2	\$44	2 (0%)	3 (0%)	4 (0%)	6 (0%)	11 (0%)
2024	35	26.2	\$52	1 (0%)	2 (0%)	3 (0%)	5 (0%)	9 (0%)
2025	33	30.8	\$62	1 (12%)	2 (0%)	3 (0%)	4 (0%)	8 (0%)
2026	20	53.5	\$107	1 (55%)	1 (0%)	1 (0%)	2 (0%)	4 (0%)

MNQ = median contracts per trade at that budget; skip = share of signals where 1 MNQ risks more than the budget.

Prop-firm evaluation

Each simulated evaluation draws random whole trading days from 2019-Aug 2026 (every trade of that day, in time order) for up to 250 trading days. It passes at the profit target and fails at the trailing drawdown (closed trades) or the daily loss limit. 5,000 runs per row. The rules are generic examples. The combined rows put H47 and H41 in the same account at the stated risks.

Account set-up	Pass	Fail	Not done in 250 days	Median days to pass
Example A: target \$3,000 · trailing DD \$2,000 · daily loss \$1,000				
H47 alone, \$150 per trade	39%	4%	58%	166
H47 alone, \$250 per trade	64%	23%	13%	108
H47 alone, \$500 per trade	56%	44%	0%	39
H47 \$150 + H41 \$150 (1 MNQ up to \$250)	58%	42%	0%	38

Account set-up	Pass	Fail	Not done in 250 days	Median days to pass
H47 \$250 + H41 \$150 (1 MNQ up to \$250)	57%	43%	0%	31
H41 alone, \$150 (1 MNQ up to \$250), for reference	55%	45%	0%	44
Example B: target \$6,000 · trailing DD \$3,000 · daily loss \$2,000				
H47 alone, \$150 per trade	4%	0%	95%	212
H47 alone, \$250 per trade	30%	8%	62%	175
H47 alone, \$500 per trade	55%	41%	4%	95
H47 \$150 + H41 \$150 (1 MNQ up to \$250)	63%	35%	2%	95
H47 \$250 + H41 \$150 (1 MNQ up to \$250)	63%	37%	1%	80
H41 alone, \$150 (1 MNQ up to \$250), for reference	56%	39%	5%	111

- **Alone at small risk, H47 is too slow.** At \$150 per trade, 58% of Example A runs (and 95% of Example B runs) had not finished after 250 trading days. It has about 33 chances a year to make money.
- **Alone at \$250 it has good odds, but it is slow.** Under Example A it passes 64% of runs, the best of any row, because its drawdowns are shallow. The median pass takes about 108 trading days, roughly five months. \$500 is faster but fails more often.
- **Together with H41 it is faster and at least as reliable.** Example A: H41 alone passes 55% (median 44 days). Adding H47 at \$150 gives 58% in 38 days. Example B: 56% for H41 alone, 63% combined. The two strategies' daily P&L correlation is only about 0.1.
- **Suggested use:** run H47 in the same account as H41 at \$150–\$250 per trade. If you can wait several months and want the best odds, H47 alone at about \$250 is a reasonable alternative. H41 pass rates here use a 250-day window, so they are higher than the 120-day figures in the H41 sizing note.
- Many firms trail the drawdown on unrealised equity, and H47 holds its runners to 15:59. The day-resampling also does not capture long bad regimes. Real pass rates will be somewhat lower.

Discussion

Strength of evidence. Over 2010–2026 it made +247R on 484 trades (+0.51R per trade, t 4.45 in R, PF 1.84 in dollars). The part that could not have influenced selection, 2023 to Aug 2026, is 119 trades at +0.51R (t 2.32 in R, PF 2.03). Both the holdout (p 0.020) and 2026 (SUPPORTED) cleared the preregistered bars.

How to read the dollar curve. On 1 NQ, dollar risk per trade grew from about \$70 (2010) to about \$1,000 (2026), so recent years dominate the dollar totals. In R, the record is spread fairly evenly across 2012–2026. The fixed-risk section shows the same record at constant dollar risk.

Where the P&L comes from. All of it comes from the ~29% of trades that are still open at 16:00 (+4.0R on average). The partial at +1R mainly softens the losers: trades that bank it and then stop out lose about half the usual amount. After the freeze the calmest third of days was roughly flat (+0.07R). The mid- and high-volatility gap days made the money (+0.84R and +0.65R). This is descriptive only.

Management. Nearly every stop/exit scheme in the library was profitable in the holdout. Schemes that cap the winners (small targets, early break-even) did much worse in 2026. The plain hold-to-close variant did as well as or better than the frozen partial plan. None of this is a reason to change the frozen rule. It does confirm the edge lives in the runners.

Portfolio fit. H47 is the natural partner for H41: different signal, different time of day (one 09:45 decision versus all-day momentum), and a correlation of about 0.1. Keep H47's own siblings (#22, #23) out of the same budget. They are the same trade (correlation 0.5–0.6).

Recommendation

Recommendation: SECOND CANDIDATE. Forward-test it next to H41, at small fixed risk, in the same account. It has the highest profit factor of the 25 frozen strategies (2.03 after the freeze) and was positive in every research stage. It had the second-strongest untouched holdout (t 2.18, null p 0.020), and 2026 was SUPPORTED. It barely moves with H41 (correlation about 0.1). The catch is sample size. It trades only about 33 times a year, so the post-freeze record is 119 trades, and a few big days carry the result. Plan for roughly half the recent edge, and expect some flat years.

Decision requested: approve H47 as the **second strategy for forward testing**, traded in the same account as H41 at small fixed risk.

Pilot design. MNQ, fixed risk per trade: contracts = risk $\div$ (ATR_h $\times$ \$2), rounded down. \$150–\$250 per trade works at current volatility with nothing skipped (details in the fixed-risk section). The partial is rounded to whole contracts, so at 1–2 MNQ everything simply runs to the close. Follow the rule exactly; one missed trade can be a quarter of the year.

Alternative: forward-paper only via `forward/run_forward.py`. With about 33 trades a year, reaching the first gate takes roughly two years, which is why a small live pilot alongside H41 is suggested.

Conditions, limits and kill criteria

- Scale-up gate: at least 60 true-forward trades (about 2 years) averaging +0.2R or better, net positive under stress costs, and no drawdown beyond 14R (the post-freeze-edge bootstrap 95th percentile). Then risk may double.
- Kill switch: a drawdown beyond 20R (the planning-case 95th percentile, above the worst since 2010), or 60 forward trades averaging below 0R.
- Regime watch: a sustained Nasdaq bear market is the untested case. If gap-up confirmations keep failing while the index trends down for two or more quarters, pause and review.
- Keep H47 and its siblings #22 and #23 in one gap-and-go budget (correlation 0.5–0.6). H41 has its own budget.
- No parameter, filter or management changes. Every observation in this report is descriptive.

Caveats

- Backtests use 1-minute bars with pessimistic same-bar handling and fixed slippage. They do not model queue position, fast-market slippage beyond 2 ticks per side (stress case), or exchange outages.

- The 2026 sample (Jan–10 Aug) is short. The campaign's own record rates it low-power (cohort null p 0.26).
- The R-based view and the parameter/management grids in this report were computed after the holdout and 2026 results were known. They are diagnostic only and must not be used to modify or re-select the strategy. The preregistered process says the only valid new evidence now comes from the true-forward record starting 24 Sep 2026.
- This is a quantitative research assessment of a systematic strategy, not personalised investment advice. Capital allocation decisions rest with the committee.

Appendix A: frozen specification (verbatim)

H47 (#4, long only)

```
{
  "costs": {
    "baseline_round_trip_usd": 14.0,
    "cost_version": "costs_v2_approved_2026-09-21#a27ed5df0c04"
  },
  "direction": "long",
  "entry_conditions": {
    "down_event": "at 09:30+15min, the RTH open gapped >= 0.25D below the previous RTH close and the close is >= 0.05D below the RTH open",
    "mode": "continuation",
    "rule": "continuation: BUY on up-event, SELL on down-event; direction=long",
    "up_event": "at 09:30+15min, the RTH open gapped >= 0.25D above the previous RTH close and the close is >= 0.05D above the RTH open"
  },
  "entry_id": "E8f8a54882d6f9ad0",
  "family": "v3_h47_overnight_same_dir_open",
  "instrument": "NQ",
  "management": {
    "abs_target": null,
    "be_R": null,
    "be_after_leg": null,
    "exit_opp": false,
    "hold": "eod",
    "id": "P25_1R_RUN",
    "kind": "partial",
    "legs": [
      [
        0.25,
        1.0
      ],
      [
        0.75,
        null
      ]
    ],
    "needs_stop": true,
    "params": {},
    "trail": null
  },
  "parameters": {
    "g": 0.25,
    "mode": "continuation",
    "open": "same",
    "w": 15
  },
  "stop": {
    "method": "atr",
    "mult": 1.0
  },
  "strategy_id": "Cfa444b4386db8d33",
  "trading_window": {
    "entry_end": "15:30",
    "entry_start": "09:30",
    "flat_at": "16:00",
    "name": "rth"
  }
}
```

Appendix B: provenance

Item	Value
Final cohort freeze	freezes/FINAL_COHORT_FREEZE.json · sha256 f5666dac728aa2539b2f8223309012914d71923ab71f3363e9e035344ace8f03
Frozen (UTC)	2026-09-22T23:21:51.429326+00:00
Discovery freeze	sha256 0343a974953d53dc026fd1f7ef5c124c8985213b132fba5df4f8a986fc85b2c9
Master record	results/V3_MASTER_RESEARCH_RECORD.md
Sealed holdout / audit trades	results/holdout/trades.parquet · results/audit2026/trades.parquet
This report	reports/H47_Gap_and_Go_Long_Research_Report.pdf, built from trades regenerated with the frozen specs (exact match with the sealed records) + stored discovery/confirmation evidence
Known erratum	results/audit2026/ERRATUM_2026-09-22.json (X3 null p only; does not affect this report)